

Software Engineering Project Report

Group 03

Software Engineering Project Report

Smart Home Automation System — Group 03

1. Group Info

Field	Details
Group Number	Group 03
Case Study	Smart Home Automation System
Course	Software Engineering
Instructor	<i>Dr.Samer Elkababji</i>

Team Members

Name	Student ID
Abdulrhman Atassi	<i>20210651</i>
Sarah Shobaki	<i>20220188</i>
Osama Zaid	<i>20220099</i>
Mariam Laith	<i>20220872</i>

2. Overview

System Purpose

The Smart Home Automation System is an IoT-based platform designed to integrate and control smart home devices including thermostats, lighting, security cameras, and alarms. The system enables homeowners and family members to monitor and manage their home environment through mobile and web applications, supporting remote device control, automation routines, and real-time alerts for sensor-triggered events. An embedded AI/ML module learns user behavior patterns to optimize energy efficiency and comfort over time.

Tools Used

Tool	Purpose
PlantUML	All UML and C4 diagram generation
Visual Studio Code	Report and source file editing
GitHub	Version control and collaboration

3. Diagrams

Part I — Context

ID	Diagram	File	Description
C4-L1	System Context Diagram	uml/c4_l1_context.puml	Shows all actors and external systems interacting with the Smart Home Automation System
C4-L2	Container Diagram	uml/c4_l2_containers.puml	Breaks the system into deployable containers: clients, API gateway, backend services, IoT gateway, and data layer
AD1	Activity Diagram — Device Control	uml/ad1_device_control.puml	Models the end-to-end flow when a user issues a device control command
AD2	Activity Diagram — Automation & AI	uml/ad2_automation.puml	Models the background automation trigger and AI suggestion flow
AD3	Activity Diagram — Onboarding	uml/ad3_onboarding.puml	Models user registration and device onboarding flow

ID	Diagram	File	Description
AD4	Activity Diagram — Alert Response	uml/ad4_alert_response.puml	Modeling incident detection and emergency response flow
AD5	Activity Diagram — Manage Family	uml/ad5_manage_family.puml	Modeling homeowner's flow for adding and managing family member accounts
AD6	Activity Diagram — Admin Monitoring	uml/ad6_admin_monitoring.puml	Modeling system administrator's monitoring and configuration flow

Part II — Interactions

ID	Diagram	File	Description
UC1	Use Case Diagram — Users	uml/uc1_user_interactions.puml	Actions available to Homeowner and Family Member
UC2	Use Case Diagram — System & Admin	uml/uc2_system_admin_interactions.puml	Actions for System Administrator, Automation Engine, and AI/ML Service
SD1	Sequence Diagram — High Level	uml/sd1_highlevel.puml	Spallholder-level view of the device control flow
SD2	Sequence Diagram — Device Control	uml/sd2_device_control.puml	Detailed developer-level view of device control including cache, DB, and IoT protocol

ID	Diagram	File	Description
SD3	Sequence Diagram — Automation & AI	uml/sd3_automation	Deleted file. Detailed description of sensor-triggered automation and AI pattern recognition.
SD4	Sequence Diagram — Onboarding	uml/sd4_onboarding	Deleted file. Detailed description of user registration and device handshake.
SD5	Sequence Diagram — Alert Response	uml/sd5_alert_detail	Deleted file. Detailed description of anomaly detection, emergency actions, and incident resolution.
SD6	Sequence Diagram — Family Member	uml/sd6_family_member	Deleted file. Detailed description of adding a family member and propagating device permissions.
SD7	Sequence Diagram — Admin	uml/sd7_admin_detail	Deleted file. Detailed description of admin system health monitoring and configuration actions.

Part III — Structure

ID	Diagram	File	Description
CD1	Class Diagram	uml/class_diagram	UML domain model with attributes, operations, generalization, aggregation, and composition.

Part IV — Behavior

ID	Diagram	File	Description
ST1	State Diagram — Device Lifecycle	uml/state_diagram_device.puml	Model defines a device can be in and the events that trigger transitions — event-driven dimension
DFD0	DFD Level 0 — Context	uml/dfd_10_context.puml	System-level data flow context showing external entities and data in/out
DFD1	DFD Level 1 — Data Pipeline	uml/dfd_11_pipeline.puml	Detailed data management pipeline covering user data, device data, AI learning, and reporting

4. Repository Structure

```

/
  README.md                                # This file - also embedded as report title page
  docs/
    report.md                              # Full project report in Markdown
    report.pdf                             # Final report converted via Pandoc
  uml/
    c4_l1_context.puml                     # C4 Level 1 - System Context
    c4_l2_container.puml                   # C4 Level 2 - Container Diagram
    ad1_device_control.puml                # Activity Diagram 1 -
Device Control
    ad2_automation_ai.puml                 # Activity Diagram 2 -
Automation & AI
    ad3_onboarding.puml                    # Activity Diagram 3 -
Onboarding
    ad4_alert_response.puml                # Activity Diagram 4 -
Alert Response
    ad5_manage_family.puml                 # Activity Diagram 5 -
Manage Family
    ad6_admin_monitoring.puml              # Activity Diagram 6 -
Admin Monitoring

```

	uc1_user_interactions.puml	# Use Case Diagram 1 -
Users		
	uc2_system_admin.puml	# Use Case Diagram 2 -
System & Admin		
	sd1_highlevel.puml	# Sequence Diagram 1 -
High Level		
	sd2_device_control_detailed.puml	# Sequence Diagram 2 -
Device Control		
	sd3_automation_detailed.puml	# Sequence Diagram 3 -
Automation & AI		
	sd4_onboarding_detailed.puml	# Sequence Diagram 4 -
Onboarding		
	sd5_alert_detailed.puml	# Sequence Diagram 5 -
Alert Response		
	sd6_family_member_detailed.puml	# Sequence Diagram 6 -
Family Member		
	sd7_admin_detailed.puml	# Sequence Diagram 7 -
Admin		
	class_diagram.puml	# Class Diagram - Full Domain Model
	state_diagram_device.puml	# State Diagram - Device Lifecycle
	dfd_l0_context.puml	# DFD Level 0 - Context
	dfd_l1_pipeline.puml	# DFD Level 1 - Data Pipeline

5. Contributions

Member Roles

Member	Role
Sarah	System Architect — C4 diagrams, container design
Abdulrhman	Interaction Modeler — Use case diagrams and descriptions
Mariam	Behavior Modeler — Activity diagrams
Osama	Sequence Modeler — Sequence diagrams and report

Commits Per Member

Member	Commits
Sarah	(32)
Abdulrhman	(18)

Member	Commits
Mariam	(1)
Osama	(1)

Table of Contents

1. System Description
 2. Part I — Context
 - 2.1 C4 Level 1: System Context Diagram
 - 2.2 C4 Level 2: Container Diagram
 - 2.3 Activity Diagrams
 3. Part II — Interactions
 - 3.1 Use Case Diagrams
 - 3.2 Use Case Descriptions
 - 3.3 Sequence Diagrams
 4. Part III — Structure
 - 4.1 Class Diagram
 5. Part IV — Behavior
 - 5.1 State Diagram
 - 5.2 Data Flow Diagram
 6. Design Decisions and Assumptions
 7. Traceability Matrix
-

1. System Description

The Smart Home Automation System is an IoT-based platform designed to integrate and centrally control smart home devices including thermostats, lighting systems, security cameras, door locks, and environmental sensors. The system serves multiple user roles within a household — a Primary Homeowner with full administrative control, and Family Members with scoped device access — alongside a System Administrator responsible for backend health and configuration.

The platform exposes its functionality through both a mobile application (iOS and Android) and a web application, allowing users to issue real-time device commands, configure automation rules, and receive instant alerts when sensor thresholds are exceeded. An embedded AI/ML module continuously analyzes per-user behavioral data to generate personalized automation suggestions and optimize energy consumption patterns without requiring manual input.

The system maintains a central database containing device configurations, user accounts, automation rules, behavioral logs, and incident records. A dedicated

IoT Gateway translates between internet-facing HTTP/REST communication and physical device protocols including Zigbee, Bluetooth, and MQTT. A Redis-based device state cache ensures sub-millisecond read latency for real-time dashboard updates and automation rule evaluation.

The system is designed around four core operational flows: device control initiated by human actors, autonomous automation triggered by sensor events, user and device onboarding, and safety-critical incident detection and response. Each flow is fully modeled across all UML layers in this report.

2. Part I — Context

2.1 C4 Level 1: System Context Diagram

The C4 Level 1 diagram presents the Smart Home Automation System as a single black box surrounded by the actors and external systems that interact with it. This level answers the question: who uses the system and what does it depend on?

Actors:

- **Primary Homeowner** — the primary human actor with full control over devices, household members, and system settings
- **Family Member** — a registered household user with permissions scoped by the Homeowner to specific devices
- **System Administrator** — a technical operator responsible for system health, device configuration, and user management

External Systems:

- **Mobile / Web Application** — the frontend interface through which human actors interact with the system
- **IoT Devices** — the physical smart devices in the household: thermostats, lights, cameras, alarms, and sensors
- **Notification Service** — an external push/email delivery platform that dispatches alerts to users
- **AI/ML Module** — a learning module that analyzes usage patterns and generates automation suggestions
- **Authentication Service** — an OAuth 2.0 identity provider that validates user credentials

Design decision: Guest User was explicitly excluded. The system is modeled as private to registered household members only. This is documented as a design assumption in Section 6.

Diagram: `uml/c4_l1_context.puml`

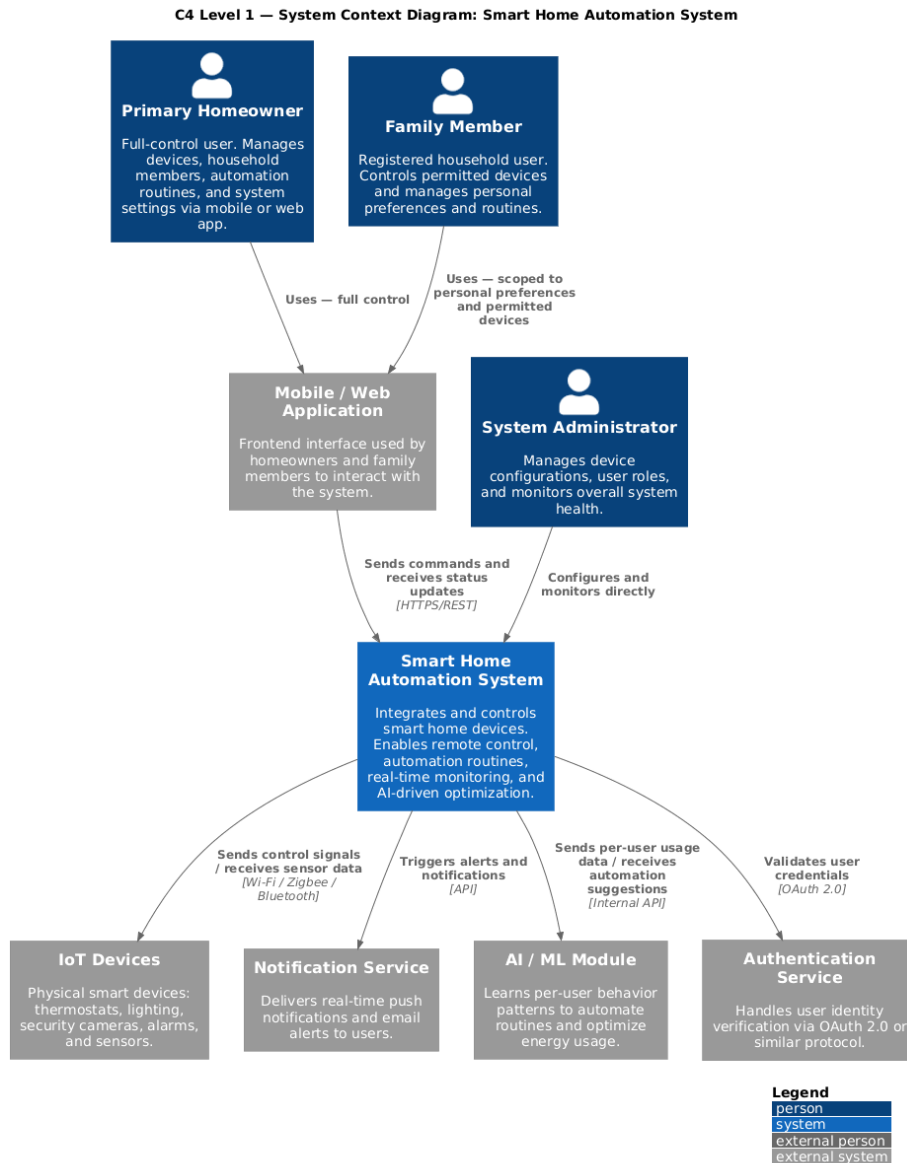


Figure 1: C4 Level 1 — System Context Diagram

2.2 C4 Level 2: Container Diagram

The C4 Level 2 diagram opens the black box and reveals the major deployable components — containers — that make up the Smart Home Automation System, along with their communication protocols.

Containers and design rationale:

Container	Type	Design Rationale
Mobile App	iOS/Android Client	Explicitly required by case study
Web Application	Browser Client	Explicitly required by case study
API Gateway	REST Entry Point	Centralizes authentication, rate limiting, and routing
User & Preference Service	Backend Service	Manages per-user profile data described in the case study
Device Management Service	Backend Service	Core domain service for device state, registration, and command dispatch
Automation Engine	Backend Service	Isolated from Device Management — rule evaluation logic is fundamentally different from device I/O
Notification Service	Backend Service	Internal dispatch layer before handing off to external push/email systems
IoT Gateway	Protocol Bridge	Required to translate between HTTP/REST and physical device protocols (Zigbee/MQTT)
AI/ML Service	ML Module	Integrated module per case study — lives inside the system boundary at L2
Central Database	PostgreSQL	Explicitly required by case study for persistent storage

Container	Type	Design Rationale
Device State Cache	Redis	Real-time device states require sub-millisecond reads — polling PostgreSQL directly is architecturally unsound

Diagram: `uml/c4_l2_container.puml`

2.3 Activity Diagrams

Six activity diagrams are provided, each modeling a distinct process flow using swimlanes to represent actor and component responsibilities. The six flows were selected to achieve complete coverage of all actors, all containers defined in C4 L2, and all primary use cases.

Activity Diagram 1 — Device Control Flow Models the primary real-time flow when a user issues a device control command. Covers the full path from user input through authentication, permission checking, IoT command dispatch, retry logic, and state persistence.

Swimlanes: Homeowner / Family Member, Mobile/Web App, API Gateway, Device Management Service, IoT Gateway

Diagram: `uml/ad1_device_control.puml`

Activity Diagram 2 — Automation and AI Trigger Flow Models the background intelligence layer initiated by a sensor event rather than a human. Covers Automation Engine rule evaluation, AI pattern analysis, and user-facing suggestion handling.

Swimlanes: IoT Gateway, Device Management Service, Automation Engine, AI/ML Service, Notification Service, Homeowner/Family Member

Diagram: `uml/ad2_automation_ai.puml`

Activity Diagram 3 — User Registration and Device Onboarding Models the system setup phase covering account creation, email verification, and the IoT Gateway handshake with a newly added physical device.

Swimlanes: New User, Mobile/Web App, API Gateway, User & Preference Service, Device Management Service, IoT Gateway

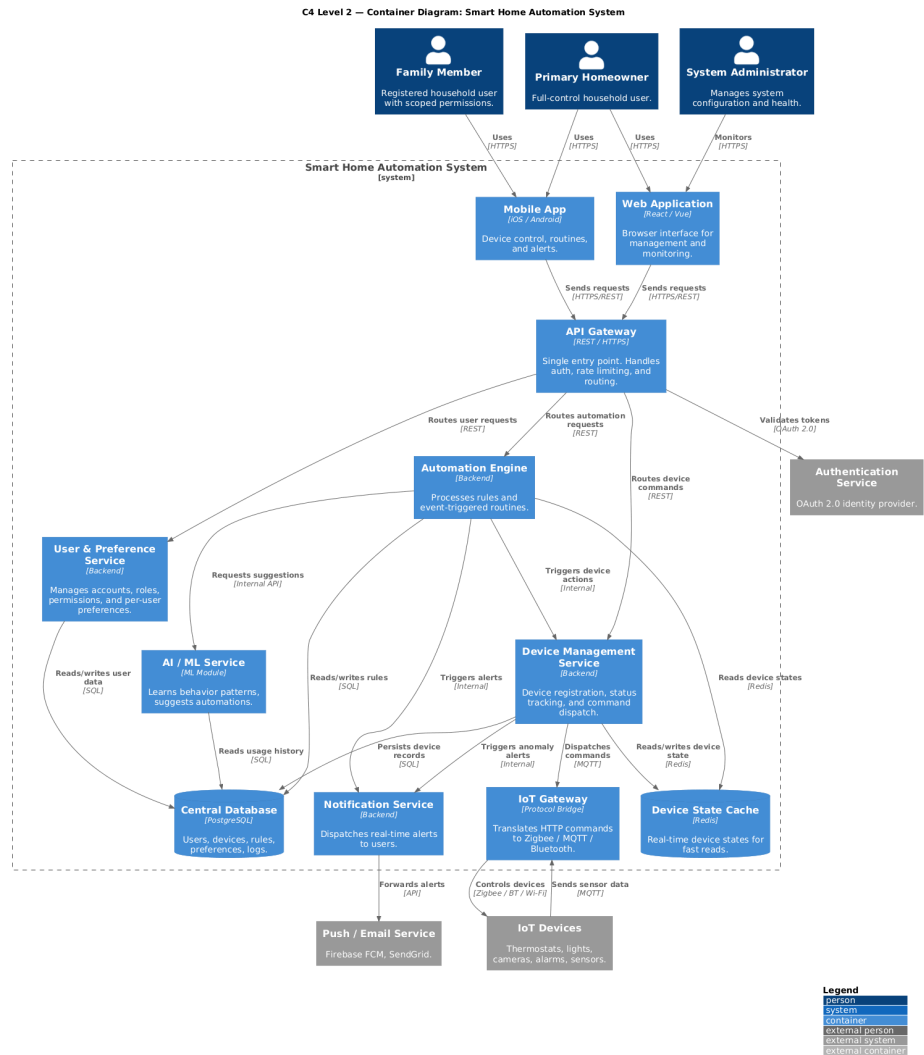


Figure 2: C4 Level 2 — Container Diagram

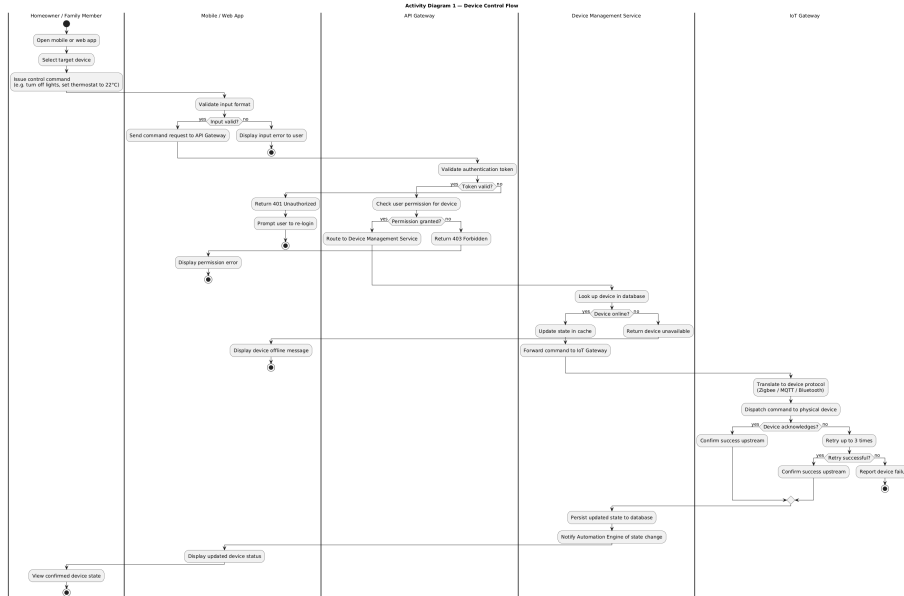


Figure 3: Activity Diagram 1 — Device Control Flow

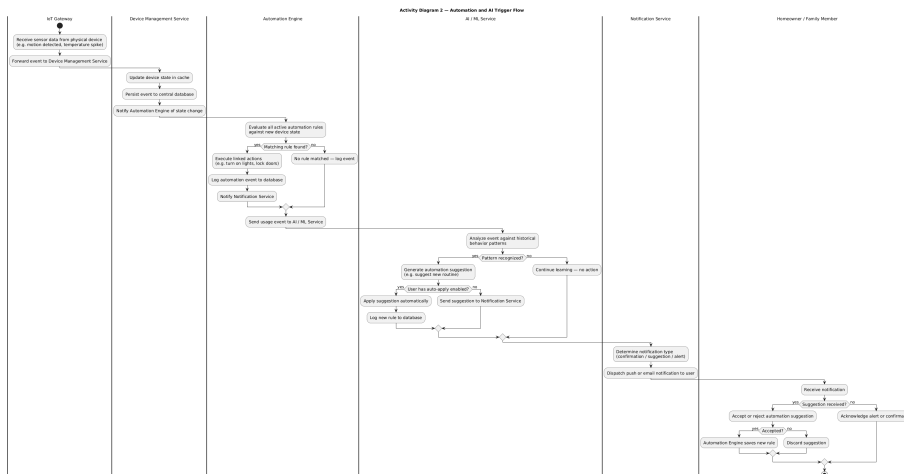


Figure 4: Activity Diagram 2 — Automation and AI Trigger Flow

Diagram: uml/ad3_onboarding.puml

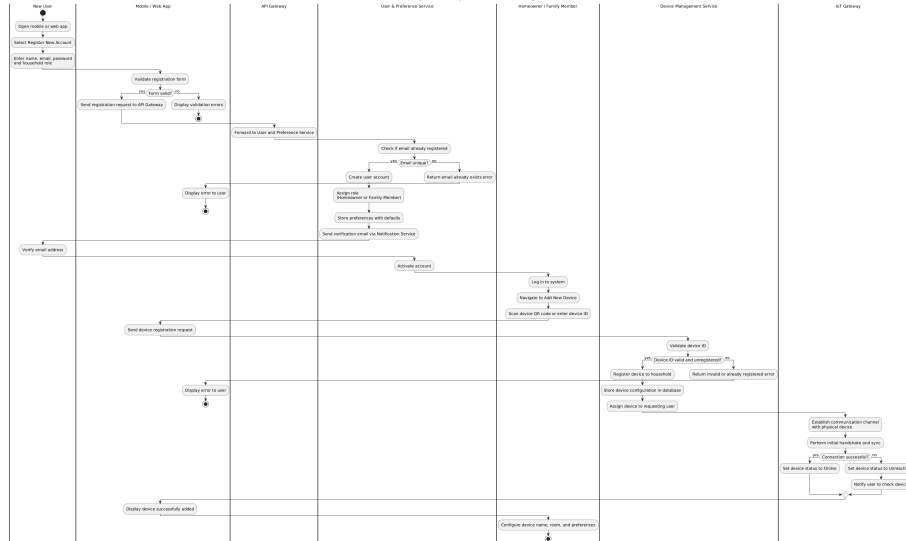


Figure 5: Activity Diagram 3 — User Registration and Device Onboarding

Activity Diagram 4 — Alert and Incident Response Flow Models the safety-critical path covering anomaly detection from a physical sensor, emergency rule execution, user notification, false alarm handling with AI feedback, and incident resolution.

Swimlanes: IoT Device/Sensor, IoT Gateway, Device Management Service, Automation Engine, Notification Service, Homeowner/Family Member, AI/ML Service

Diagram: uml/ad4_alert_response.puml

Activity Diagram 5 — Manage Family Members Models the Homeowner's flow for adding a Family Member. This flow is architecturally significant because it involves permission propagation across the Device Management Service — not a trivial CRUD operation.

Swimlanes: Homeowner, Mobile/Web App, API Gateway, User & Preference Service, Device Management Service, Notification Service, New Family Member

Diagram: uml/ad5_manage_family.puml

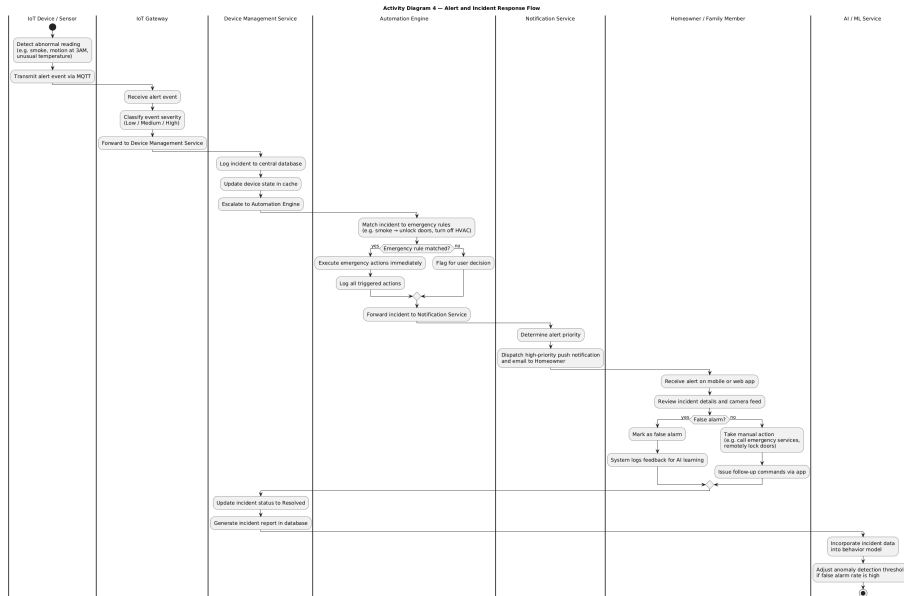


Figure 6: Activity Diagram 4 — Alert and Incident Response Flow

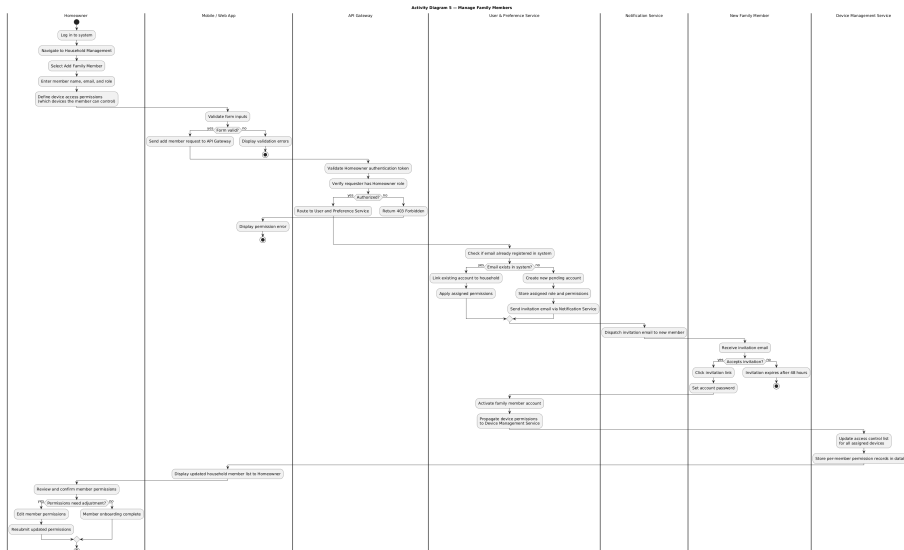


Figure 7: Activity Diagram 5 — Manage Family Members

Activity Diagram 6 — System Administrator Monitoring and Configuration Models the System Administrator’s interaction pattern — no IoT layer involved, direct backend service access, administrative operations rather than device-driven flows.

Swimlanes: System Administrator, Web Application, API Gateway, Device Management Service, User & Preference Service, Automation Engine

Diagram: `uml/ad6_admin_monitoring.puml`

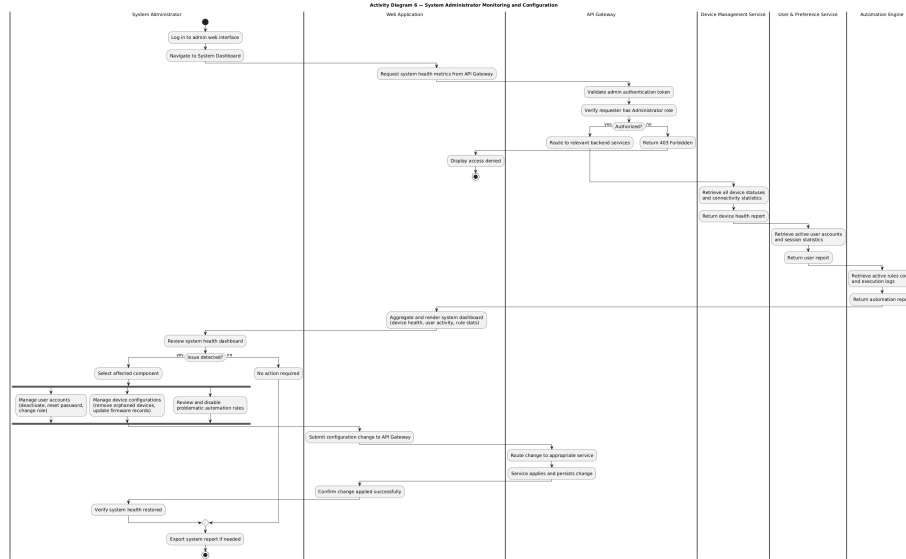


Figure 8: Activity Diagram 6 — System Administrator Monitoring

3. Part II — Interactions

3.1 Use Case Diagrams

Two use case diagrams are provided. The split separates human-actor use cases from system-actor and administrative use cases, preventing the diagram from becoming unreadable.

Actor-to-package associations are used in both diagrams for diagram clarity. Each actor interacts with all use cases within their associated package. This decision is explicitly documented here and in the diagram notes.

Use Case Diagram 1 — User Interactions Covers all use cases accessible to the Homeowner and Family Member. The Homeowner has access to

all packages. The Family Member is restricted to Devices, Monitoring, and Preferences.

Diagram: `uml/uc1_user_interactions.puml`

Use Case Diagram 2 — System and Admin Interactions Covers use cases triggered by the System Administrator, Automation Engine, and AI/ML Service.

Diagram: `uml/uc2_system_admin.puml`

3.2 Use Case Descriptions

Detailed tabular descriptions are provided for the eight primary use cases — one per major functional area.

UC-01: Log In

Field	Description
Use Case ID	UC-01
Use Case Name	Log In
Actor(s)	Homeowner, Family Member
Description	A registered user authenticates into the system using their credentials to gain access to permitted features
Preconditions	User has a registered and verified account
Postconditions	User is authenticated and redirected to the system dashboard
Main Flow	1. User opens mobile or web app. 2. User enters email and password. 3. App sends credentials to API Gateway. 4. API Gateway forwards to Authentication Service. 5. Authentication Service validates credentials and returns JWT token. 6. System grants access based on user role

Field	Description
Alternate Flow	3a. Invalid credentials → system displays error and prompts retry. 3b. Account not verified → system prompts email verification
Exceptions	Account locked after 5 failed attempts. Network failure → system displays connection error
Includes	—
Extends	—

UC-02: Control Device

Field	Description
Use Case ID	UC-02
Use Case Name	Control Device
Actor(s)	Homeowner, Family Member
Description	An authenticated user sends a control command to a smart device
Preconditions	User is logged in. Target device is registered and online. User has permission to control the target device
Postconditions	Device executes the command. State is updated in cache and database. User receives confirmation
Main Flow	1. User selects target device. 2. User issues control command. 3. App sends request to API Gateway. 4. Gateway validates authentication and permission. 5. Device Management Service forwards command to IoT Gateway. 6. IoT Gateway translates and dispatches to physical device. 7. Device acknowledges command. 8. State updated in cache and database.
Alternate Flow	9. Confirmation sent to user 7a. Device does not acknowledge → IoT Gateway retries up to 3 times. 7b. All retries fail → system reports device failure and triggers alert

Field	Description
Exceptions	Device offline → system displays offline message. Permission denied → system returns 403 error
Includes	UC-01 Log In
Extends	UC-06 Trigger Alert

UC-03: Add / Remove Device

Field	Description
Use Case ID	UC-03
Use Case Name	Add / Remove Device
Actor(s)	Homeowner
Description	The homeowner registers a new smart device to the household or removes an existing one
Preconditions	User is logged in as Homeowner. For adding: device is physically powered and within communication range
Postconditions	Device is registered/removed in the central database. IoT Gateway establishes or terminates communication channel
Main Flow	1. Homeowner navigates to device management. 2. Selects Add Device. 3. Scans QR code or enters device ID. 4. Device Management Service validates ID. 5. System registers device to household. 6. IoT Gateway performs handshake with physical device. 7. Device status set to Online. 8. Homeowner assigns name, room, and preferences
Alternate Flow	4a. Device ID already registered → system returns conflict error. 6a. Handshake fails → device set to Unreachable, user notified
Exceptions	Invalid device ID → system rejects registration. Network failure during handshake → system prompts retry
Includes	UC-01 Log In
Extends	—

Field	Description
-------	-------------

UC-04: Create Automation Rule

Field	Description
Use Case ID	UC-04
Use Case Name	Create Automation Rule
Actor(s)	Homeowner
Description	The homeowner defines a trigger-based automation rule that instructs the system to perform device actions automatically when specified conditions are met
Preconditions	User is logged in as Homeowner. At least one device is registered
Postconditions	Automation rule is saved to database. Automation Engine begins evaluating rule against incoming events
Main Flow	1. Homeowner navigates to automation settings. 2. Selects Create New Rule. 3. Defines trigger condition. 4. Defines action. 5. Saves rule. 6. System stores rule in database. 7. Automation Engine registers rule for evaluation
Alternate Flow	5a. Rule conflicts with existing rule → system warns user and requests confirmation
Exceptions	Invalid trigger or action configuration → system displays validation error
Includes	UC-01 Log In
Extends	UC-08 Generate Automation Suggestion

UC-05: Manage Family Members

Field	Description
Use Case ID	UC-05
Use Case Name	Manage Family Members

Field	Description
Actor(s)	Homeowner
Description	The homeowner adds, removes, or modifies household member accounts and assigns device access permissions
Preconditions	User is logged in as Homeowner
Postconditions	Family member account is created, updated, or removed. Permissions are propagated to the Device Management Service
Main Flow	1. Homeowner navigates to household management. 2. Selects Add Member. 3. Enters member name and email. 4. Assigns role and device permissions. 5. System creates pending account and sends invitation email. 6. Member accepts invitation and sets password. 7. Account activated with assigned permissions. 8. Permissions propagated to Device Management Service
Alternate Flow	3a. Email already registered → system links existing account to household. 6a. Invitation not accepted within 48 hours → invitation expires
Exceptions	Maximum household member limit reached → system rejects addition
Includes	UC-01 Log In
Extends	—

UC-06: Trigger Alert

Field	Description
Use Case ID	UC-06
Use Case Name	Trigger Alert
Actor(s)	Automation Engine
Description	The Automation Engine detects an anomalous device event and dispatches a high-priority alert to the homeowner

Field	Description
Preconditions	A sensor event or device failure has occurred. Notification Service is operational
Postconditions	Alert is delivered to homeowner via push notification and/or email. Incident is logged to database
Main Flow	1. Automation Engine detects anomalous event. 2. Classifies alert severity. 3. Forwards to Notification Service. 4. Notification Service dispatches push and email alert. 5. Homeowner acknowledges alert. 6. Incident logged to database
Alternate Flow	4a. Push notification fails → system falls back to email only
Exceptions	Notification Service unavailable → incident logged locally and alert queued for retry
Includes	—
Extends	UC-02 Control Device

UC-07: Execute Automation Rule

Field	Description
Use Case ID	UC-07
Use Case Name	Execute Automation Rule
Actor(s)	Automation Engine
Description	The Automation Engine evaluates active rules against incoming device state changes and executes matched actions automatically
Preconditions	At least one automation rule exists in the database. A device state change event has been received
Postconditions	Matched actions are executed. Execution is logged. User is notified if rule specifies notification

Field	Description
Main Flow	1. Device state change received. 2. Automation Engine loads active rules. 3. Evaluates each rule against current state. 4. Identifies matching rules. 5. Dispatches action commands to Device Management Service. 6. Logs execution to database. 7. Sends confirmation notification if configured
Alternate Flow	4a. No rule matches → engine logs event and takes no action
Exceptions	Rule references a deleted device → engine skips rule and flags it for review
Includes	—
Extends	—

UC-08: Generate Automation Suggestion

Field	Description
Use Case ID	UC-08
Use Case Name	Generate Automation Suggestion
Actor(s)	AI / ML Service
Description	The AI/ML Service analyzes historical user behavior and device usage patterns to generate personalized automation rule suggestions
Preconditions	Sufficient usage history exists in the database. AI/ML Service is operational
Postconditions	Suggestion is either auto-applied if enabled, or sent to homeowner for review

Field	Description
Main Flow	1. AI/ML Service reads usage history. 2. Identifies recurring behavior patterns. 3. Generates automation suggestion. 4. Checks if homeowner has auto-apply enabled. 5a. If yes → submits rule to Automation Engine. 5b. If no → sends suggestion to Notification Service. 6. Homeowner accepts or rejects suggestion
Alternate Flow	3a. No significant pattern detected → service logs attempt and schedules next analysis
Exceptions	Insufficient data → suggestion generation skipped until minimum data threshold is met
Includes	—
Extends	UC-04 Create Automation Rule

3.3 Sequence Diagrams

Seven sequence diagrams are provided — one high-level stakeholder view and six detailed developer-level diagrams. Each detailed diagram traces directly back to its corresponding activity diagram.

Traceability Map

Activity Diagram	Sequence Diagram	Primary Use Case(s)
AD1 — Device Control	SD1 (high-level) + SD2 (detailed)	UC-02 Control Device
AD2 — Automation & AI	SD3 (detailed)	UC-07 Execute Automation Rule, UC-08 AI Suggestion
AD3 — Onboarding	SD4 (detailed)	UC-01 Log In, UC-03 Add Device
AD4 — Alert Response	SD5 (detailed)	UC-06 Trigger Alert
AD5 — Manage Family	SD6 (detailed)	UC-05 Manage Family Members
AD6 — Admin Monitoring	SD7 (detailed)	Monitor System Health, Manage User Accounts

Use Case Diagram 1 — User Interactions

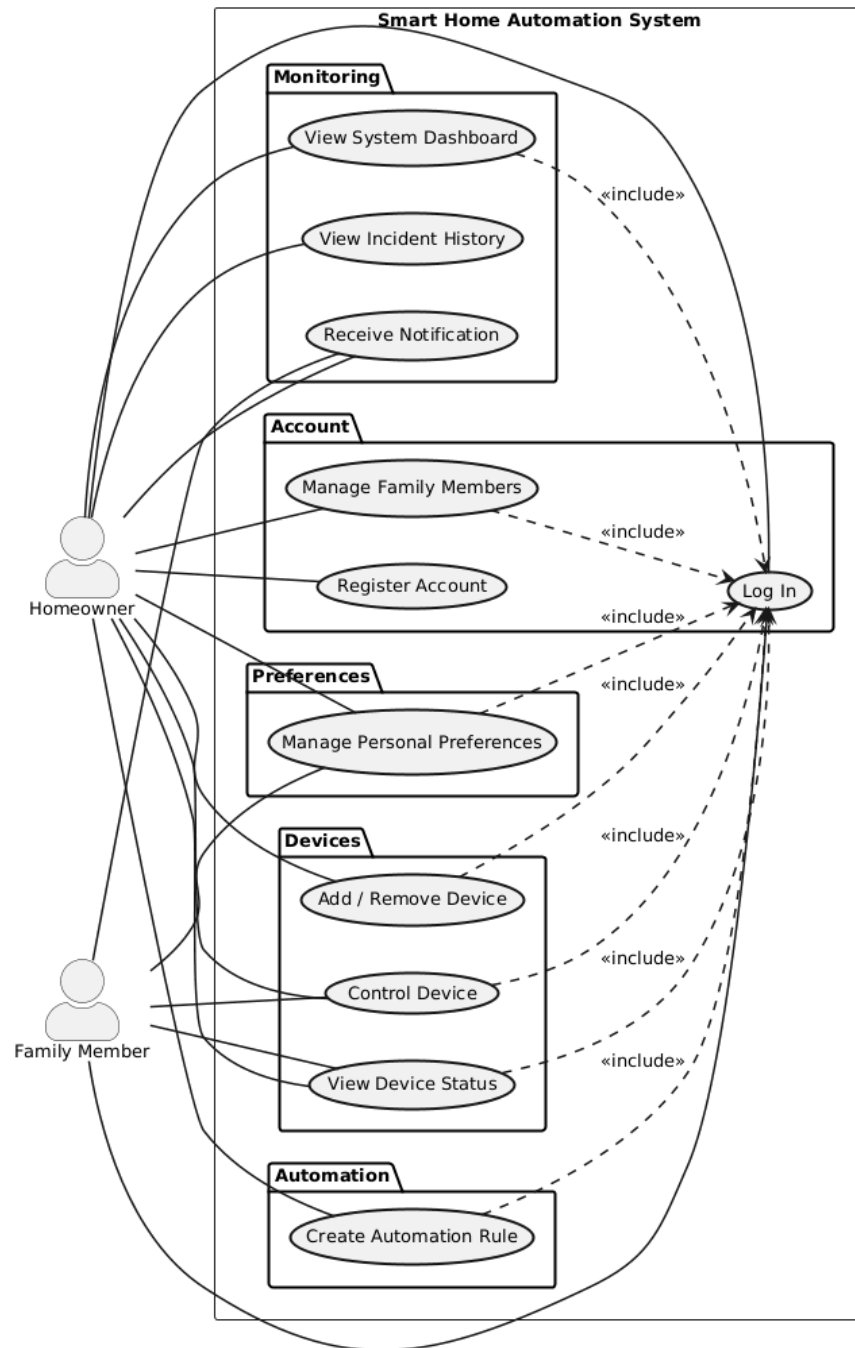


Figure 9: Use Case Diagram 1 — User Interactions

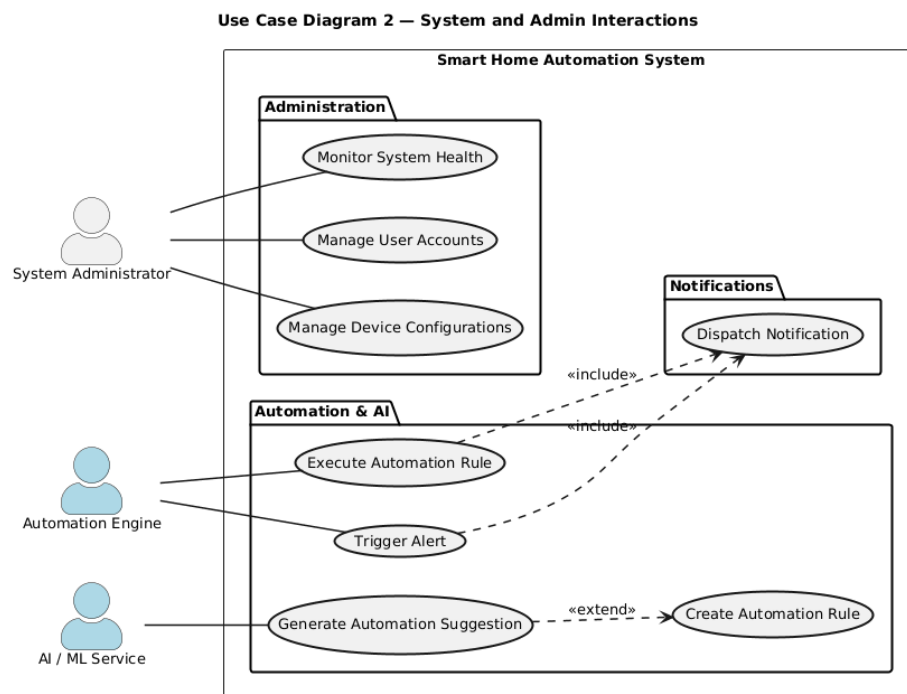


Figure 10: Use Case Diagram 2 — System and Admin Interactions

Sequence Diagram 1 — Device Control Flow (High-Level / Stakeholder View) Diagram: uml/sd1_highlevel.puml

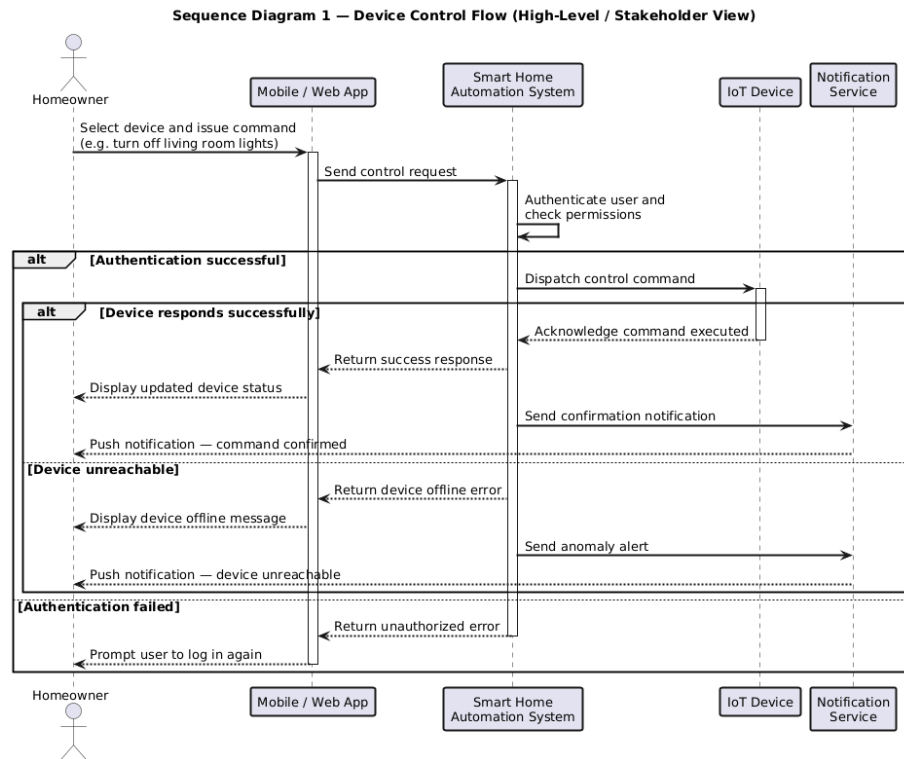


Figure 11: Sequence Diagram 1 — High-Level Device Control

Sequence Diagram 2 — Device Control Flow (Detailed / Developer View) Shows JWT validation, Redis cache reads and writes, Zigbee frame dispatch, HTTP status codes, and Automation Engine notification on state change.

Key design decision: Cache is read before the database is written because the cache reflects the most current live state. The database write happens only after the IoT device acknowledges the command.

Diagram: uml/sd2_device_control_detailed.puml

Sequence Diagram 3 — Automation and AI Trigger Flow (Detailed) Models the fully autonomous background flow. No human initiates this sequence

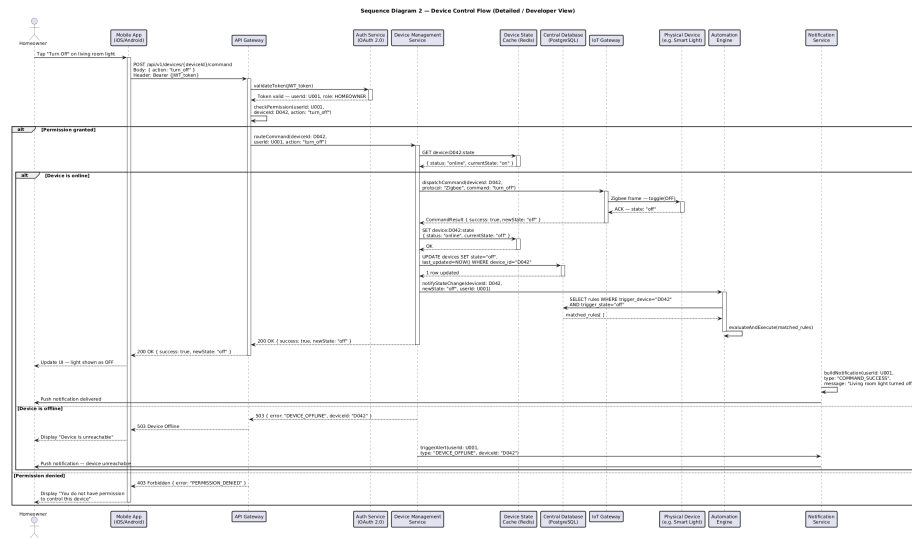


Figure 12: Sequence Diagram 2 — Detailed Device Control

— a physical sensor triggers the entire chain.

Diagram: uml/sd3_automation_detailed.puml

Sequence Diagram 4 — User Registration and Device Onboarding (Detailed) Covers the full lifecycle from registration form submission through email verification to IoT device handshake.

Diagram: uml/sd4_onboarding_detailed.puml

Sequence Diagram 5 — Alert and Incident Response Flow (Detailed) Models the safety-critical path including emergency rule execution, false alarm handling, and AI threshold recalibration.

Diagram: uml/sd5_alert_detailed.puml

Sequence Diagram 6 — Manage Family Members (Detailed) Models the permission propagation chain from User & Preference Service to Device Management Service access control lists.

Diagram: uml/sd6_family_member_detailed.puml

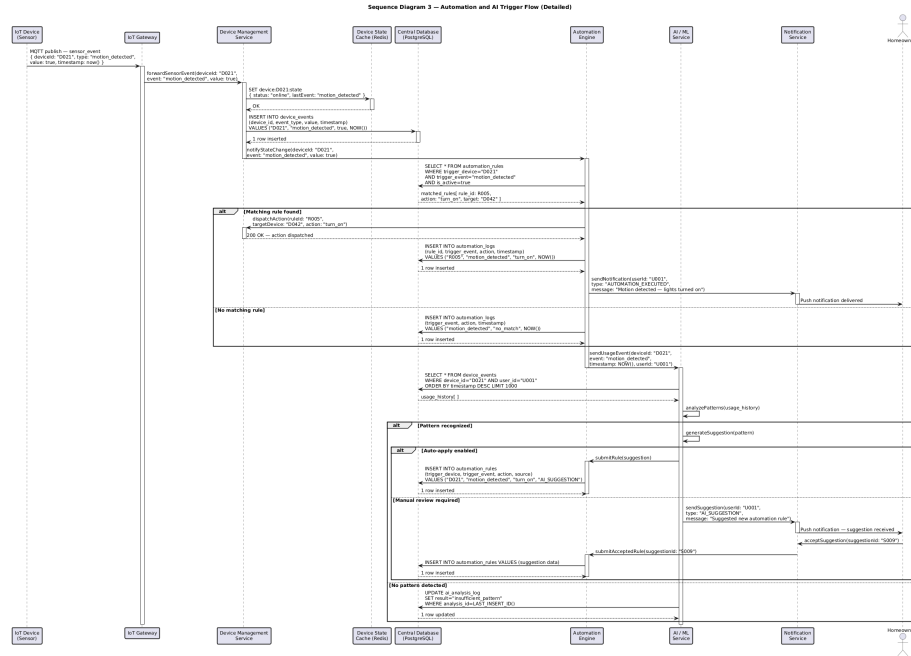


Figure 13: Sequence Diagram 3 — Detailed Automation and AI

Sequence Diagram 7 — System Administrator Monitoring and Configuration (Detailed) Models a fundamentally different access pattern — no IoT layer, direct backend service access via admin-only endpoints.

Diagram: uml/sd7_admin_detailed.puml

4. Part III — Structure

4.1 Class Diagram

The class diagram models the static structure of the Smart Home Automation System — all domain entities, their attributes and operations, and the relationships between them. Every class was derived directly from entities that appeared in the sequence and activity diagrams, ensuring full consistency across all UML layers.

Classes and their source:

Class	Derived From
User (abstract)	All SDs — base class for all human actors
Homeowner	C4 L1, UC1, AD1–AD6

Class	Derived From
FamilyMember	C4 L1, UC1, AD5, SD6
Administrator	C4 L1, UC2, AD6, SD7
Household	SD4, SD6
Device	All SDs — core domain entity
DeviceState	SD2, SD3, SD5
Permission	SD6 — member_permissions table
AutomationRule	AD2, SD3
AutomationLog	SD3, SD7
AIModel	AD2, SD3
AI suggestion	SD3 — suggestionId: S009
Incident	AD4, SD5
Notification	All SDs
Preference	AD5, SD6
Invitation	AD5, SD6

Relationships:

Relationship Type	Between
Generalization (Inheritance)	User — Homeowner, FamilyMember, Administrator
Composition	Household — Device, User — Preference, Device — DeviceState, AutomationRule — AutomationLog, FamilyMember — Permission
Aggregation	Household — User
Association	AutomationRule → Device, Incident → Device, Notification → User, AIModel → AI suggestion, AI suggestion → AutomationRule, Permission → Device, Homeowner → Invitation, Invitation → FamilyMember

Why User is abstract: You never instantiate a plain User — every person in the system is either a Homeowner, FamilyMember, or Administrator. Making it abstract enforces this constraint and correctly models the inheritance hierarchy.

Why Composition for Household — Device: A Device cannot exist independently — it must belong to a Household. If the Household is deleted, its devices are deleted too.

Why Aggregation for Household — User: A User can exist in the system independently of a Household — they have an account before accepting an

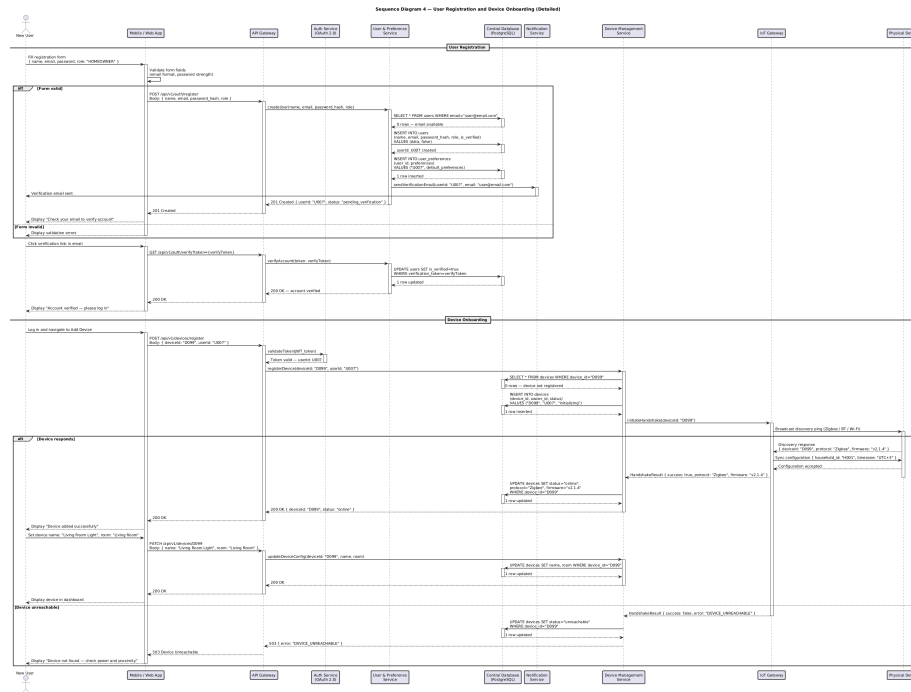


Figure 14: Sequence Diagram 4 — Detailed Onboarding

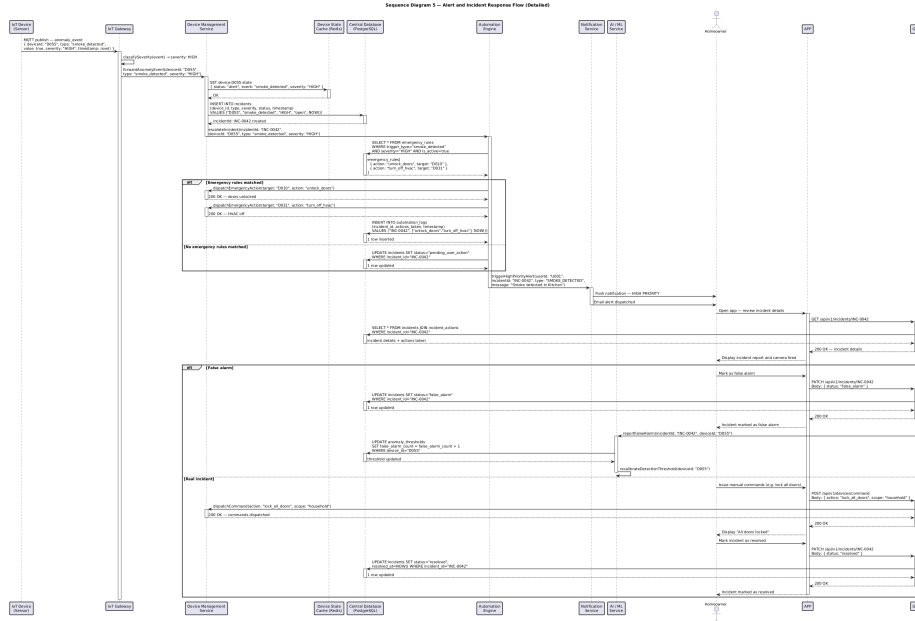


Figure 15: Sequence Diagram 5 — Detailed Alert Response

invitation. The parts can outlive the whole.

Why AISuggestion is a separate class from AutomationRule: In SD3 we explicitly modeled a suggestion object with its own ID that the user accepts or rejects. It only becomes an AutomationRule after acceptance — they are two distinct entities.

Diagram: uml/class_diagram.puml

5. Part IV — Behavior

The Smart Home Automation System is classified as a **hybrid system** — it exhibits both event-driven and data-driven behavior:

- **Event-driven dimension:** IoT sensors fire events, devices transition between states, the Automation Engine reacts autonomously to those events
- **Data-driven dimension:** User data, device configuration, and usage history flow through the system and are processed, stored, and transformed into outputs like reports and AI suggestions

Accordingly, two behavioral models are provided: a State Diagram for the event-driven dimension, and a Data Flow Diagram for the data-driven dimension.

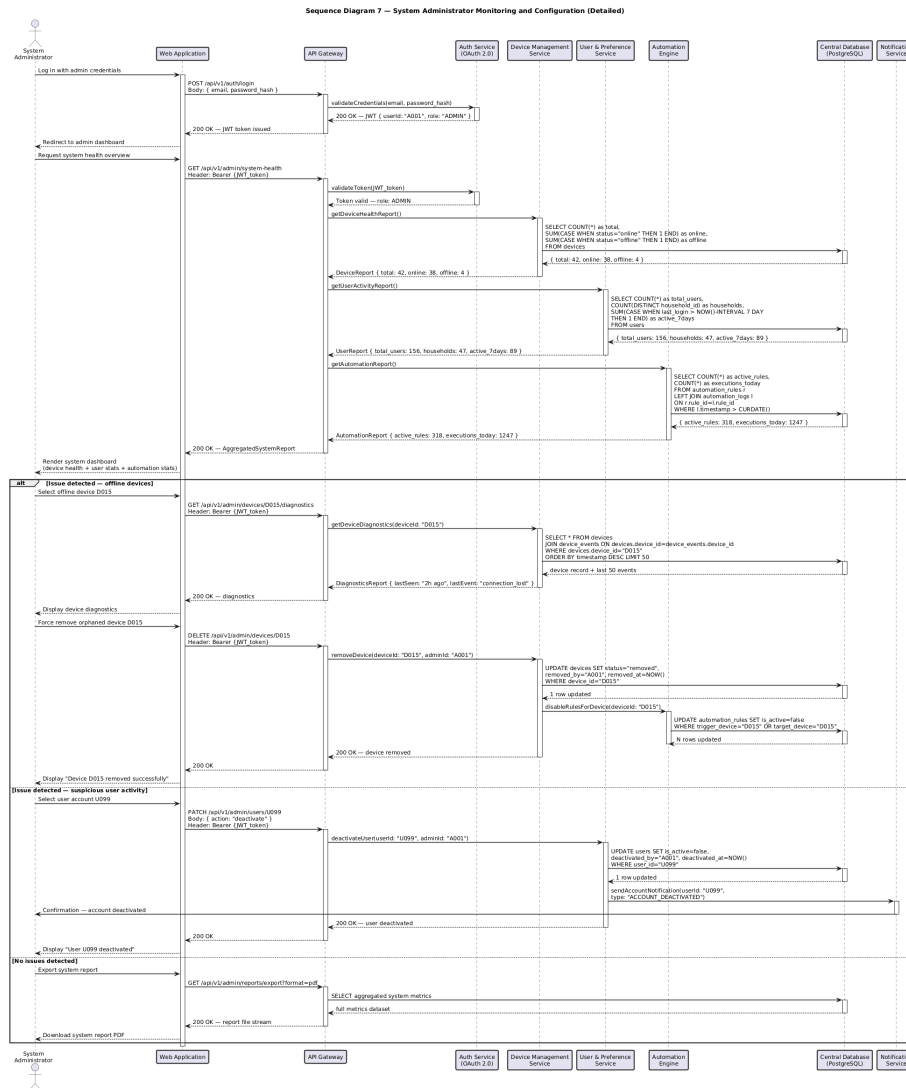


Figure 17: Sequence Diagram 7 — Detailed Admin Monitoring

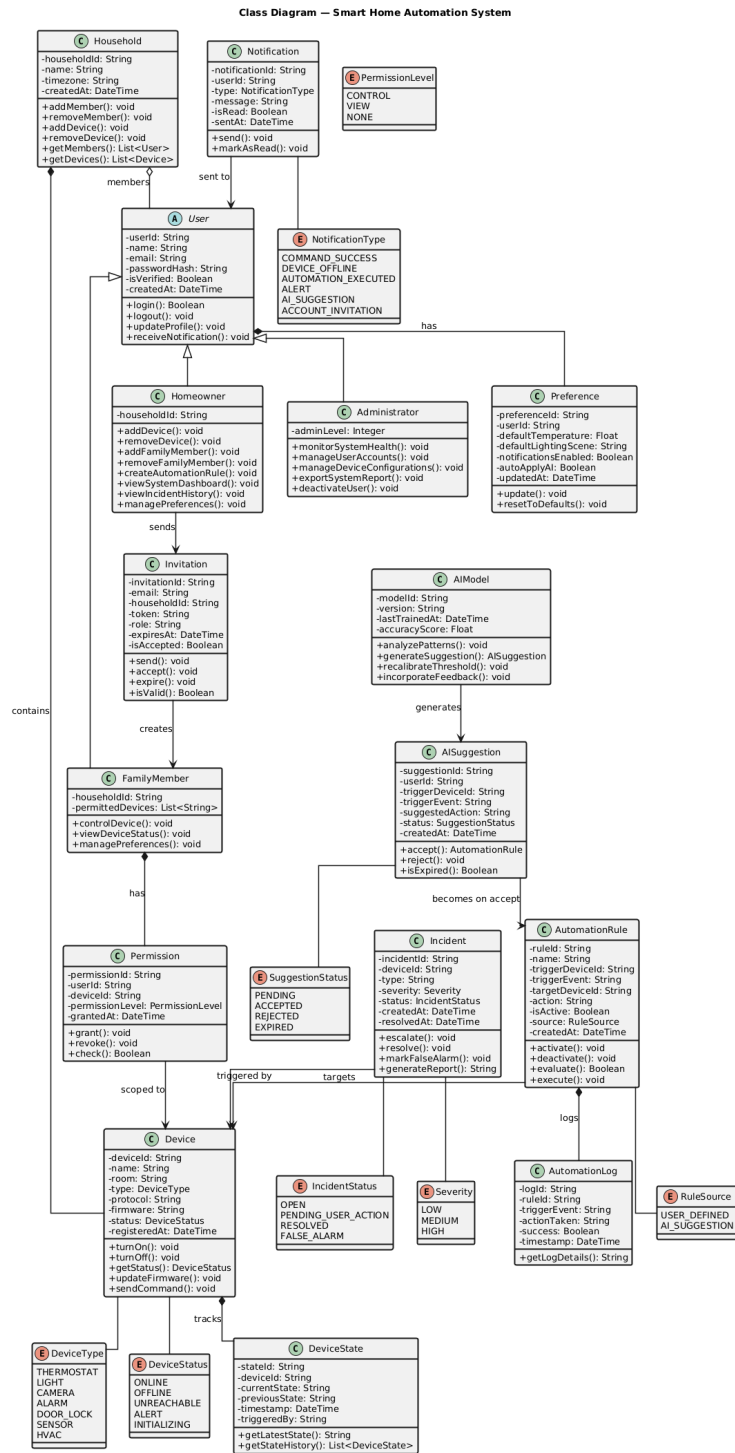


Figure 18: Class Diagram — Smart Home Automation System

5.1 State Diagram — Device Lifecycle

The state diagram models the complete lifecycle of a Device entity — the most architecturally significant entity in the system because it is at the center of every operational flow.

States:

State	Description
Unregistered	Device exists physically but has not been added to the system
Initializing	Device is being onboarded — IoT Gateway handshake in progress
Online	Device is registered and connected. Contains two substates: Idle and Executing
Idle	Device is online and ready — no active command
Executing	Device is actively processing a dispatched command
Unreachable	Device is registered but not responding to communication
Alert	Device has detected an anomaly or triggered an incident
Maintenance	Device has been flagged for inspection or firmware update
Removed	Device has been deregistered — terminal state

Why Online is a composite state: `Online` contains `Idle` and `Executing` as substates. This is the correct UML modeling because these two substates only exist in the context of a connected device — they have no meaning outside `Online`.

Why Alert transitions to Maintenance and not directly to Removed: A safety-critical system should never automatically remove a device that is generating alerts — the sensor itself may need physical inspection. A human decision is required before removal, enforced by routing through `Maintenance` first.

Why Removed is terminal: Once deregistered, all state history and rules are archived. Re-adding the same physical device starts a brand new lifecycle from `Unregistered`.

Diagram: `uml/state_diagram_device.puml`

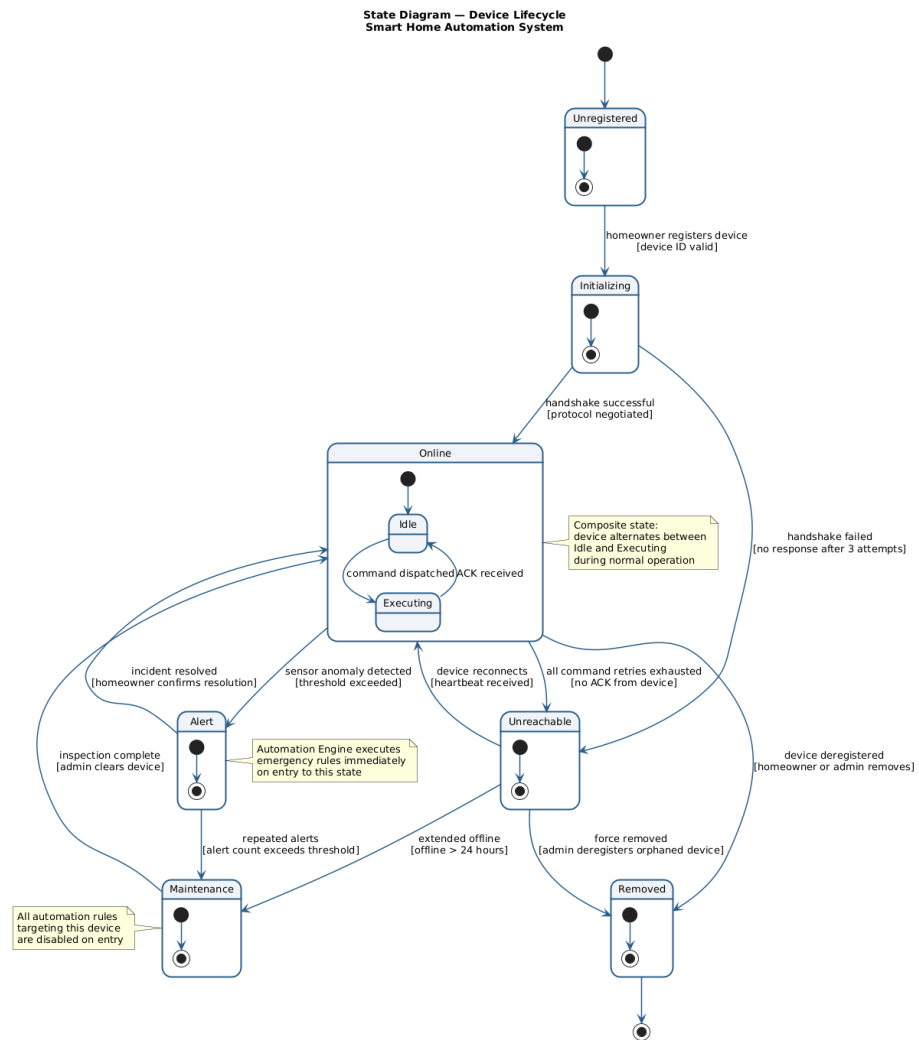


Figure 19: State Diagram — Device Lifecycle

State-Stimulus Table

Current State	Stimulus / Event	Next State	Action Taken
Unregistered	Homeowner registers device ID	Initializing	IoT Gateway initiates handshake
Initializing	Handshake successful	Online	Device status set to Online in DB and cache
Initializing	Handshake failed	Unreachable	Device status set to Unreachable, user notified
Online / Idle	Command dispatched	Online / Executing	Device Management Service forwards command to IoT Gateway
Online / Executing	ACK received from device	Online / Idle	State updated in cache and DB, user notified
Online / Executing	All retries exhausted	Unreachable	Failure reported, anomaly alert triggered
Online	Sensor anomaly detected	Alert	Automation Engine executes emergency rules, alert dispatched
Online	Admin or Homeowner deregisters	Removed	Device record soft-deleted, rules disabled
Alert	Homeowner resolves incident	Online	Incident marked resolved, AI feedback logged
Alert	Homeowner marks false alarm	Online	Incident marked false alarm, AI threshold recalibrated
Alert	Alert count exceeds threshold	Maintenance	Device flagged, rules disabled, admin notified
Unreachable	Heartbeat received	Online	Device status updated in cache and DB
Unreachable	Offline > 24 hours	Maintenance	Device flagged, admin notified

Current State	Stimulus / Event	Next State	Action Taken
Unreachable	Admin force removes	Removed	Device record soft-deleted, rules disabled
Maintenance	Admin clears device	Online	Device status restored, rules re-enabled
Removed	—	—	Terminal state — no further transitions

5.2 Data Flow Diagram

The DFD models the data-driven dimension of the system — how data enters, gets processed, gets stored, and produces outputs. The DFD intentionally excludes event-driven flows (device commands, automation triggers, incident response) which are already covered by the State Diagram and Activity Diagrams.

The DFD focuses exclusively on the **data management pipeline**: user and household data, device configuration data, AI learning data, and system reporting.

DFD Level 0 — Context Shows the entire data pipeline as a single process with external entities and data flows in and out.

External Entities:

- **Homeowner / Family Member** — provides registration data, preferences, profile updates; receives personalized responses, AI suggestions, usage reports
- **IoT Devices** — provides sensor readings, device telemetry, configuration data; receives configuration sync and firmware updates
- **System Administrator** — provides management requests; receives system health reports and analytics

Diagram: `uml/dfd_10_context.puml`

DFD Level 1 — Data Management Pipeline Breaks the system into four focused data management processes:

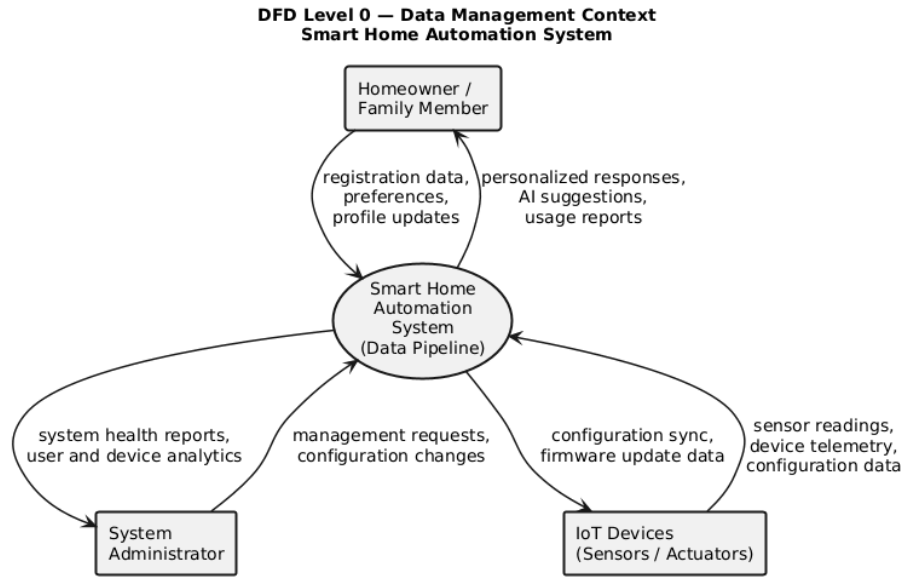


Figure 20: DFD Level 0 — Data Management Context

Process	Description
P1 — Manage User & Household Data	Handles registration, profile updates, preferences, and family member management
P2 — Manage Device & Configuration Data	Handles device registration, telemetry storage, configuration sync, and firmware tracking
P3 — AI Learning Data Pipeline	Reads usage history and preferences, trains the AI model, generates and stores automation suggestions
P4 — Generate Reports & Dashboards	Aggregates data from all stores and produces usage dashboards and system health reports

Data Stores:

Store	Contents
D1 — Users & Preferences	User accounts, roles, preferences, notification settings
D2 — Devices & Configurations	Device records, protocols, firmware versions, room assignments

Store	Contents
D3 — Usage History & Events	Processed behavior patterns and device event history
D4 — AI Model & Suggestions	Trained model data and generated automation suggestions
D5 — Reports & Logs	Generated system reports and audit logs

Why Device State Cache is excluded from the DFD: The Redis cache is a real-time infrastructure component that supports the event-driven command flow — it is not part of the data management pipeline. It is correctly represented in C4 L2 as a separate container and in the sequence diagrams as a participant.

Diagram: `uml/dfd_11_pipeline.puml`

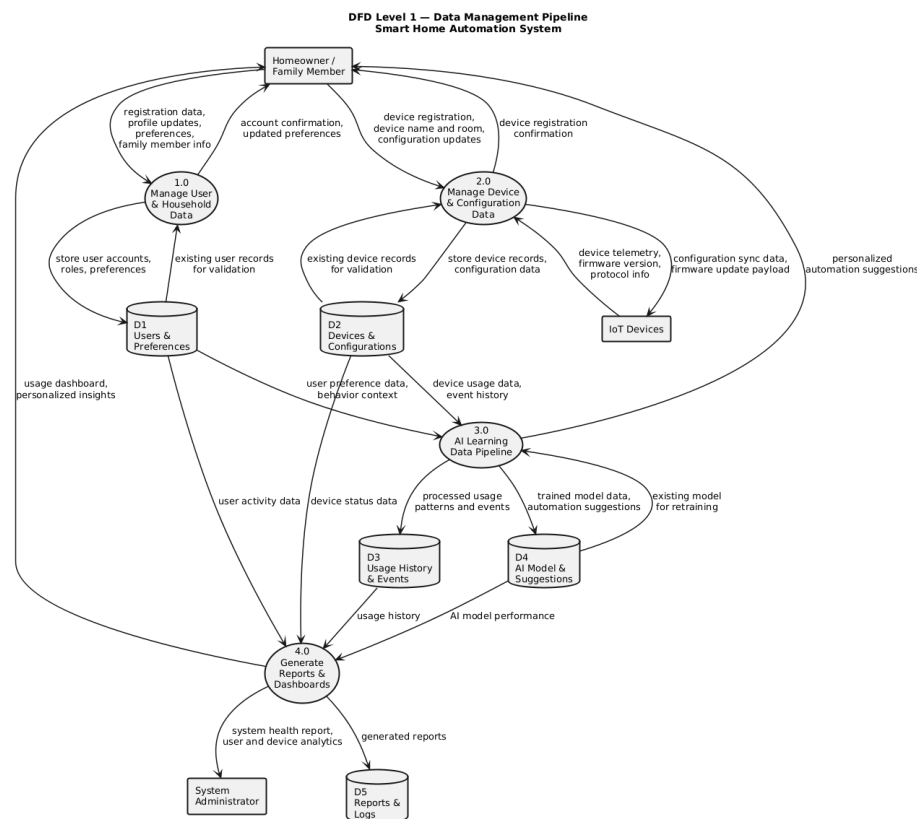


Figure 21: DFD Level 1 — Data Management Pipeline

6. Design Decisions and Assumptions

#	Decision / Assumption	Justification
1	Guest User excluded from actor model	The system is modeled as private to registered household members only
2	Family Member defined as a distinct actor from Homeowner	Case study mentions per-user preferences — implies individual profiles with different permission scopes
3	AI/ML Module treated as external at C4 L1 and internal at C4 L2	At L1 the system is a black box. At L2 the case study describes AI as an integrated module
4	IoT Gateway modeled as a separate container	Physical devices speak Zigbee/MQTT; the backend speaks HTTP/REST. Protocol translation is a distinct architectural concern
5	Redis cache introduced for device state	Real-time dashboards require sub-millisecond state reads. Polling PostgreSQL directly is architecturally unsound for IoT systems
6	Actor-to-package associations used in use case diagrams	With 17 use cases and 5 actors, individual lines produce an unreadable diagram. Package-level associations are a documented professional practice
7	Six activity diagrams provided instead of one	Complete actor coverage requires separate diagrams for Family Member (AD5) and System Administrator (AD6) flows

#	Decision / Assumption	Justification
8	System classified as hybrid	The system has both significant event-driven behavior (IoT sensors, device state transitions) and data-driven behavior (AI pipeline, reporting)
9	DFD scoped to data management pipeline only	Event-driven flows are already covered by State Diagram and Activity Diagrams. Mixing them into the DFD would produce an inaccurate and unreadable diagram
10	Device State Cache excluded from DFD	Redis is a real-time infrastructure component supporting the event-driven layer — not part of the data management pipeline modeled by the DFD
11	User is an abstract class	No plain User is ever instantiated — every person is a Homeowner, FamilyMember, or Administrator
12	AI Suggestion modeled as separate class from AutomationRule	A suggestion has its own lifecycle and ID — it only becomes a rule after user acceptance

7. Traceability Matrix

Actor Coverage

Actor	C4 L1	C4 L2	Activity Diagram	Use Case Diagram	Sequence Diagram	Class Diagram
Homeowner			AD1,3,4,5,6	UC1	SD1,2,4,5,6	
Family Member			AD5	UC1	SD6	

Actor	C4 L1	C4 L2	Activity Diagram	Use Case Diagram	Sequence Diagram	Class Diagram
System Administrator			AD6	UC2	SD7	
Automation Engine			AD2,4	UC2	SD3,5	—
AI/ML Service	—		AD2,4	UC2	SD3,5	AIModel

Container Coverage

Container	Activity Diagram	Sequence Diagram
Mobile / Web App	AD1,3,5,6	SD1,2,4,5,6,7
API Gateway	AD1,3,5,6	SD2,4,5,6,7
User & Preference Service	AD3,5	SD4,6,7
Device Management Service	AD1,2,3,4,5	SD2,3,4,5,6,7
Automation Engine	AD2,4,6	SD2,3,5,7
Notification Service	AD2,3,4,5	SD2,3,4,5,6,7
IoT Gateway	AD1,2,3,4	SD2,3,4,5
AI/ML Service	AD2,4	SD3,5
Central Database	All ADs	All SDs
Device State Cache	AD1,2,4	SD2,3,5

Use Case Coverage

Use Case	Activity Diagram	Sequence Diagram	Class Diagram
UC-01 Log In	AD3	SD4	User hierarchy
UC-02 Control Device	AD1	SD1, SD2	Device, DeviceState
UC-03 Add/Remove Device	AD3	SD4	Device, Household
UC-04 Create Automation Rule	AD2	SD3	AutomationRule
UC-05 Manage Family Members	AD5	SD6	FamilyMember, Permission, Invitation
UC-06 Trigger Alert	AD4	SD5	Incident, Notification

Use Case	Activity Diagram	Sequence Diagram	Class Diagram
UC-07 Execute Automation Rule	AD2	SD3	AutomationRule, AutomationLog
UC-08 Generate AI Suggestion	AD2	SD3	AIModel, AISuggestion

Behavior Coverage

Diagram	Behavioral Dimension	Traces To
State Diagram — Device	Event-driven	AD1, AD4, SD2, SD5
DFD L0	Data-driven context	C4 L1
DFD L1	Data-driven pipeline	AD3, AD5, SD4, SD6, SD7